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Hamza Mohammed Hassanain

Software Engineer

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EDUCATION

B.Sc. Computer and Information Sciences, Suez Canal University
AWS Certified Solutions Architect – Associate, Amazon Web Services

2022 – 2026
credly.com/verify

EXPERIENCE

Founding Software Engineer Nov 2025 – Present
Repovive (YC F’26) Full Time – United States, Remote
Repovive is a science expert community where experts solve unsolved scientific problems, starting with theoretical computer science and math proof reasoning. We also host live programming and math contests, and have grown to 10,000+ users.

- Joined as the **first engineer** and built the core systems from the early days through YC.
- Led design of the distributed judge (judge1 on **Judge0**); scaled submissions **10x (50 to 500+ concurrent)**.
- Cut infrastructure cost **95% (\$600 to \$25/mo)** by migrating resource-heavy tasks from serverless to Docker/Nginx/Redis.
- Isolated **20+ languages** in resource-limited Docker images; redesigned the DB layer for **50MB+** contest test cases.
- Built AI voice interviews (**LiveKit**): recruiter, behavioral, and coding agents run from versioned, admin-authored definitions.
- Moved CI to **Vercel** when Actions quota ran out: one fail-fast pipeline (format, typecheck, test, build), no prod secrets.
- Tracked down live-contest outages with **Claude Code**, **MCP servers**, **PostHog**, and log analysis, then shipped the fixes.
- Built a real-time collaborative dev space using **WebSockets** and **Yjs CRDTs** over a Monaco live-syncing editor.

OPEN SOURCE CONTRIBUTIONS

- **GitLab Go SDK (client-go)**: typed group integrations – Slack, Jira, MS Teams, Mattermost, Harbor, +3 | **!2692, !2691, !2675, !2670**.
- **GitLab Terraform provider**: new Harbor & Mattermost integration resources + group SAML attribute | **!2881, !2909, !2852**.
- **GitLab monorepo**: `Json.safe_parse` hardening and Vue lint compliance | **!221163, !221158**.
- **LLVM / Clang**: `constexpr X86 cvtpd2ps`, `AArch64/ARM CLS` → `ISD`: `:CTLS`, `LLDB SWIG` tests | **#169980, #178885, #178430, #173473**.
- **Apple Swift NIO**: idempotent `NIOFileSystem` directory creation | **#3410**.

PROJECTS

Autonomous AI Tutor & Curriculum Engine, Graduation Project github.com/sigma-loop
• Led backend & DevOps for an 8-person team building an on-demand AI course, lesson, and challenge generator.
• Graded code in a Judge0 sandbox and math proofs via a confidence-scored LLM; owned all CI/CD and AWS deployments.

Competitive-Programming Judge — AWS Reference Architecture github.com/HamzaHassanain/CP-Judge-AWS
• Designed an AWS reference architecture: Judge0 `isolate` on ECS-on-EC2, Fargate orchestrator, Spot ASG on queue depth.
• Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

Polyman CLI npmjs.com/package/polyman-cli | github.com/HamzaHassanain/polyman
• Built a TypeScript CLI automating the Codeforces Polygon workflow: test generation, validation, and direct API sync.

Systems in C++ — HALlocator & cppress HALlocator | cppress
• Built a Red-Black-Tree best-fit memory allocator ($O(\log n)$) and an epoll-driven C++ web framework, both from scratch.

TECHNICAL WRITING

- Authored “What Every Programmer Should Know About Memory”, a series on CPU caches, virtual memory, and optimization.
- Published “I Built a Backend Web Framework from Scratch in C++”, covering non-blocking I/O, HTTP parsing, and thread pools.

ACHIEVEMENTS

ICPC ECPC & ACPC — Competitive Programming icpc.global/YC7Z02WQCPIB
17th of 500 teams at ECPC (2024); two-time ACPC regional qualifier (2024, 2025), top 250 in the Africa & Arab region.

SKILLS

Cloud & DevOps	AWS, Docker, Nginx, Redis, Terraform, CI/CD (GitHub Actions), DigitalOcean, Linux
Backend & Full-Stack	TypeScript, Node.js, Express.js, Next.js, React, MongoDB, PostgreSQL, REST APIs, WebSockets, Yjs (CRDTs)
Languages & Systems	C++, Go, Rust, Python, SQL; System Design, Distributed Systems, Message Queues